

HCl and Cl^-, H_2O and H_3O^+ are two conjugate acid – base pairs. i.e., Cl^- is the conjugate base of the acid HCl. (or) HCl is conjugate acid of Cl^-. Similarly H_3O^+ is the conjugate acid of H_2O.

Limitations of Lowry – Bronsted theory

i. Substances like BF_3, $AlCl_3$ etc., that do not donate protons are known to behave as acids.

Evaluate yourself – 2

Write a balanced equation for the dissociation of the following in water and identify the conjugate acid –base pairs.

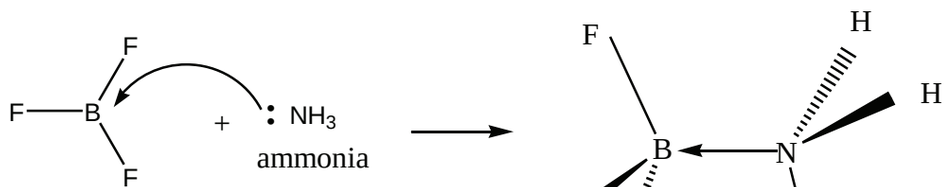
i) NH_4^+ ii) H_2SO_4 iii) CH_3COOH.

8.1.3 Lewis concept

In 1923, Gilbert . N. Lewis proposed a more generalised concept of acids and bases. He considered the electron pair to define a species as an acid (or) a base. According to him, an acid is a species that accepts an electron pair while base is a species that donates an electron pair. We call such species as Lewis acids and bases.

A Lewis acid is a positive ion (or) an electron deficient molecule and a Lewis base is a anion (or) neutral molecule with at least one lone pair of electrons.

Les us consider the reaction between Boron tri fluoride and ammonia



electron deficient - electron pair donor - adduct

(Lewis acid) - (Lewis base)

Here, boron has a vacant 2p orbital to accept the lone pair of electrons donated by ammonia to form a new coordinate covalent bond. We have already learnt that in coordination compounds, the Ligands act as a Lewis base and the central metal atom or ion that accepts a pair of electrons from the ligand behaves as a Lewis acid.

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